

AI-Powered Candidate Ranking System

Project Overview

A Transformer-based recruitment platform that ranks candidates semantically rather than relying on keyword matching. The system analyzes resumes, skills, projects, education, and experience to generate explainable candidate rankings.

Problem Statement

- Understand job requirements from a Job Description.
- Understand candidate profiles beyond keywords.
- Measure semantic compatibility between candidates and roles.
- Rank candidates based on suitability.
- Provide explainable recommendations.

Core Features

1. Semantic Candidate Matching
2. Transformer-Based Ranking
3. Skill Gap Analysis
4. Experience Evaluation
5. Project Relevance Detection
6. Explainable AI Recommendations
7. Fast Candidate Search using FAISS

Technology Stack

Programming Language: Python

Deep Learning Framework: PyTorch

NLP Models: Sentence-BERT (SBERT), Cross Encoder, MiniLM, Transformers

Vector Database: FAISS

Backend: FastAPI

Frontend: Streamlit

Data Processing: Pandas, NumPy

NLP Libraries: spaCy, NLTK

Resume Parsing: PyPDF2, pdfplumber

Architecture

Job Description → SBERT Embeddings → FAISS Retrieval → Top Candidates → Cross Encoder Re-ranking → Hybrid Scoring → Explainability Module → Final Ranking

Hybrid Scoring Formula

Final Score = $0.50 \times \text{Semantic Match} + 0.20 \times \text{Skill Match} + 0.15 \times \text{Experience Score} + 0.15 \times \text{Project Relevance}$